

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm (New) Examination
University Theory Examination April/May 2015
MA141.1 Advanced Mathematics

Date: 20/05/2015, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION I is 30 minutes. Answers to SECTION I must be written during the first 30 minutes of the examination.
3. Answers to SECTION II and SECTION III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice.

[20]

1. The value of $\binom{5}{4}$ is

(A) 9

(B) 5

(C) 4

(D) 1

2. 57th derivative of function $y = e^x + 5$ is

(A) e^x

(B) $e^x + 57$

(C) $5e^{5x}$

(D) $5e^{57x}$

3. 4th derivative of function $y = \sin x$ is

(A) $\cos x$

(B) $-\cos x$

(C) $-\sin x$

(D) $\sin x$

4. 9th derivative of function $y = 3x^4 - x^2 - 1$ is

(A) 0

(B) $27x^3 - 9$

(C) $12x^2$

(D) 12

5. Which of the following condition is not required to be satisfy by function $f(x)$ for Rolle's theorem on $[a, b]$?

(A) $f(x)$ is continuous $[a, b]$

(C) $f(a) = f(b)$

(B) $f(x)$ is differentiable on (a, b)

(D) $f'(a) = f'(b)$

6. Let $f(x)$ be a function defined on $[a, b]$ satisfying the following conditions.

(a) $f(x)$ is continuous on $[a, b]$

(b) $f(x)$ is differentiable on (a, b) . Then which of the following is true?

- (A) there exist at least $c \in [a, b]$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.
 (B) there exist at least $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.
 (C) there exist at least $c \in (a, b)$ such that $f'(c) = 0$.
 (D) there exist at least $c \in (a, b)$ such that $f'(c) = f(b) - f(a)$.

7. The Laplace transform of 1 is

- (A) $\frac{1}{s^2}$ (B) $\frac{1}{s}$ (C) 1 (D) s

8. The Laplace transform of e^{2t} is

- (A) $\frac{1}{s^2 - 1}$ (B) $\frac{1}{s^2 + 1}$ (C) $\frac{1}{s + 2}$ (D) $\frac{1}{s - 2}$

9. The inverse Laplace transform of $\frac{2}{s^3}$ is

- (A) 1 (B) t (C) t^2 (D) e^t

10. $\mathcal{L}^{-1}[f(t) - g(t)] = \dots$

- (A) $\mathcal{L}^{-1}[f(t)] + \mathcal{L}^{-1}[g(t)]$ (C) $\mathcal{L}^{-1}[f(t)]\mathcal{L}^{-1}[g(t)]$
 (B) $\mathcal{L}^{-1}[f(t)] - \mathcal{L}^{-1}[g(t)]$ (D) None of Above

11. The general form of first order linear differential equation is

- (A) $\frac{dy}{dx} + P(y)y = Q(y)$ (C) $\frac{dy}{dx} + P(x)y = Q(x)$
 (B) $x\frac{dy}{dx} + P(y)y = Q(y)$ (D) $\frac{dy}{dx} + P(x)y = Q(x)y$

12. The order and degree of differential equation $\frac{d^7y}{dx^7} - y = 0$ are

- (A) order=1 and degree=7 (C) order=7 and degree=7
 (B) order=7 and degree=1 (D) order=1 and degree=1

13. The solution of equation $\frac{dx}{dy} = 1$ is

- (A) $x = y + C$ (C) $x = 1$
 (B) $x = -y + C$ (D) $y = 1$

14. If $y = ax + b$, then differential equation satisfied by y is

(A) $\frac{d^2y}{dx^2} = 0$

(C) $\frac{d^2y}{dx^2} = b$

(B) $\frac{d^2y}{dx^2} = a$

(D) $\frac{d^2y}{dx^2} = a + b$

15. If -2 and 5 are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is

(A) $y = C_1e^{-2x} + C_2e^{5x}$

(C) $y = C_1e^{-2x} + C_2e^{-5x}$

(B) $y = C_1e^{2x} + C_2e^{-5x}$

(D) $y = C_1e^{2x} + C_2e^{5x}$

16. The particular integral of differential equation $\frac{d^2y}{dx^2} - y = 0$ is

(A) $y_p = C_1e^{-x} + C_2e^x$

(C) $y_p = C_1 \cos x + C_2 \sin x$

(B) $y_p = C_1e^{-2x} + C_2e^{2x}$

(D) $y_p = 0$

17. If $5 + 8i$ and $5 - 8i$ are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is

(A) $y = e^{8x}(C_1 \cos 5x + C_2 \sin 5x)$

(C) $y = C_1e^{5x} + C_2e^{8x}$

(B) $y = e^{5x}(C_1 \cos 8x + C_2 \sin 8x)$

(D) $y = C_1e^{-5x} + C_2e^{-8x}$

18. The Wronskian of two functions $y_1 = e^{2x}$ and $y_2 = e^{-2x}$ is

(A) 2

(B) -2

(C) 4

(D) -4

19. The governing differential equation of Newton's law of cooling is

(A) $\frac{d^2T}{dt^2} = k(T - T_A)$

(C) $\frac{dT}{dt} = 0$

(B) $\frac{dT}{dt} = k(T - T_A)$

(D) None of above

20. The governing differential equation of radio active decay model is

(A) $\frac{dn(t)}{dt} = -kn(t)$

(C) $\frac{dm^2(t)}{dt^2} = -km(t)$

(B) $\frac{dm(t)}{dt} = 0$

(D) None of above

SECTION II

Q.2. Attempt any **Four** of the following. [20]

A Find n^{th} derivative of function $y = x \log(3x + 2)$ using Leibnitz's theorem. [05]

B Solve the differential equation $\frac{dy}{dt} - y = 1$ with initial condition $y(0) = 0$ using Laplace transform. [05]

C Verify Cauchy's mean value theorem for $f(x) = x^3 - 2$ and $g(x) = x$, $x \in [5, 8]$. [05]

D Solve the differential equation $x \frac{dy}{dx} + \frac{y^2}{x} = y$. [05]

E Solve the differential equation $(D^2 - 9)y = e^{3x}$. [05]

F A pot of liquid is put on the stove to boil. The temperature of the liquid reaches 170° and then the pot is taken off the burner and placed on a counter in the kitchen. The temperature of the air in the kitchen is $76^\circ F$. After two minutes the temperature of the liquid in the pot is $123^\circ F$. How long before the temperature of the liquid in the pot will be $84^\circ F$? [05]

SECTION III

Q.3. Attempt any **Four** of the following. [20]

A 1. 7^{th} derivative of function $y = e^{3x}$ [02]

2. Find 9^{th} derivative of function $y = \sin(2x - 1)$. [03]

B 1. Find Laplace transform of function $e^{3t}t^2$. [03]

2. Find inverse Laplace transform of function $\frac{4}{(s-7)^2+16}$ [02]

C Find Taylor's series expansion of $f(x) = x^2 + x + 1$ around $a = 5$. [05]

D Solve the differential equation $\frac{dy}{dx} + y = e^{-x}$. [05]

E Solve the differential equation $\frac{d^2y}{dx^2} + y = 0$ [05]

F State the governing differential equation of population growth and decay model and solve it. [05]

Q.4. Attempt any **Four** of the following. [20]

A 1. Find 3rd derivative of function $y = e^{3x} \sin(4x + 5)$. [03]

2. Find 4th derivative of function $y = 3^{13x}$. [02]

B Solve the differential equation $(y + e^{-x}) dx + (x + 5) dy = 0$ [05]

C 1. Solve the differential equation $(D^2 - 36)y = 0$. [03]

2. Find the Wronskian of two functions $y_1 = \cos x$ and $y_2 = \sin x$. [02]

D Find value of c of the Lagrange's mean value theorem for $f(x) = x^2$, $x \in [0, 5]$. [05]

E A fossilized bone is found to contain one thousandth the original amount of C-14. Determine the age of fossil. The half life of C - 14 is approximately 5600 years. [05]

F Find inverse Laplace transform of function $\frac{1}{s(s^2 + 64)}$ [05]

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